

Preprocessing & Engineering Report

Machine Learning for Gunshot Residue Classification: Accuracy, Interpretability, and Failure Analysis

Preprocessing & Feature Engineering Report

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Executive Summary

This report presents the results of 3 different feature engineering approaches based on the specific target model (logistic regression, XGBoost, Neural Network). With the support of our primary funder, the National Institute of Justice, our findings aim to directly improve reliability and efficiency of GSR classification techniques for forensic science professionals in crime laboratories and law enforcement agencies, as well as metrology bodies like NIST who define benchmarks and standards for particle analysis. Additionally, public defenders and legal organizations such as the Innocence Project may find benefits from the results of our research, namely with false positive analysis and model limitations.

1. Background and Research Question

The identification of GSR particles is a cornerstone of forensic investigations involving firearm discharge. When a firearm is fired, microscopic particles containing characteristic elements such as lead (Pb), barium (Ba), and antimony (Sb) are deposited on nearby surfaces and can serve as physical evidence in criminal proceedings. Traditional GSR analysis relies on scanning electron microscopy with energy dispersive X-ray spectroscopy (SEM/EDS), a process that depends heavily on expert judgment and is susceptible to both false positives from chemically similar environmental particles and false negatives from atypical GSR compositions.

Research Question

How accurately and reliably can machine learning models distinguish true gunshot residue particles from chemically similar non-GSR particles? We further investigate how classification performance varies across model types (linear, tree-based, neural network), which elemental features drive predictions, and under what conditions models fail, particularly regarding false positives where environmental particles are incorrectly labeled as GSR.

2. Hypothesis & Prediction

We hypothesize that machine learning models can effectively distinguish true GSR particles from environmental confounders, with non-linear models (e.g., gradient-

boosted trees, neural networks) outperforming linear models (e.g., logistic regression) due to the complex interactions among elemental features. We further hypothesize that particles lacking the classic Pb-Ba-Sb signature will be more prone to misclassification as false negatives, while chemically similar confounders will drive higher false positive rates.

We predict that feature importance analyses (e.g., SHAP, permutation importance) will identify lead, barium, and antimony, along with their ratios, as the most influential predictors, and that misclassifications will concentrate among particles with elemental profiles that partially overlap with the Pb-Ba-Sb signature.

3. Methods

3.1. Preprocessing Pipeline

The initial preprocessing plan involved preparing a high-dimensional particle-level dataset for supervised machine learning classification. Following exploratory data analysis (EDA), several adjustments were made to better reflect the structure of the data and minimize potential modeling biases.

The preprocessing pipeline, shared across all model tracks, was applied before any feature engineering. Starting from the full NFI dataset of 2,801,667 particles across 95 columns, three steps were performed. First, the 62 non-informative elemental columns (those with non-zero values in fewer than 1% of particles) were dropped, retaining only

the 27 informative elements (O, S, Cu, Ba, Al, Si, Ca, Pb, Sb, Fe, Zn, Cl, K, Na, Mg, Ti, Sn, P, Mn, As, Cr, Br, Mo, Sr, Ni, W, and Hg). Second, the 506,682 ambiguous particles (single-element classes Pb, Ba, Sb, Sr) were removed, as these cannot be reliably assigned to either GSR or Non-GSR based on current evidence. Third, a binary target column was encoded (GSR = 1, Non_GSR = 0). The resulting preprocessed dataset contains 2,294,985 particles across 27 elemental features, with a class balance of 47.0% GSR to 53.0% Non-GSR.

No imputation was required because zero values in the elemental columns represent genuine measurements (element below detection threshold), not missing data. No outlier removal was performed because extreme elemental concentrations are physically meaningful in this domain. A particle with 90% lead is not an error but a real chemical signature.

The dataset consists of over 2.8 million observations, each representing a unique particle defined by a combination of `stub_id` and `particle_id`. Initial checks confirmed that all observations are unique at this level, eliminating concerns about duplicate rows. However, EDA revealed a more subtle issue where multiple particles originate from the same stub, introducing the risk that particles from the same source could appear in both training and testing datasets.

To address this, preprocessing incorporated a group-aware data splitting strategy using Scikit-learn's `GroupShuffleSplit`—based on the `stub_id`. This ensures that all particles

from a given stub are assigned exclusively to either the training or testing set, preventing leakage of stub-specific patterns into model evaluation. Without this approach, the model could artificially inflate performance by learning relationships tied to individual stubs rather than generalizable particle characteristics.

Additional preprocessing steps included handling missing values by removing observations with missing target labels and standardizing continuous variables using scaling techniques. Feature scaling was necessary due to the wide range of elemental measurements, ensuring that no single variable disproportionately influenced model training. These steps align with standard preprocessing practices designed to improve model stability and interpretability.

3.2. Feature Selection & Engineering Approach

Feature engineering focused on transforming raw elemental measurements into more meaningful representations that better reflect the underlying chemistry of gunshot residue geared towards how each model works under the hood. Feature engineering is a critical step in machine learning, as it improves model performance by constructing more informative inputs from raw data.

The primary feature engineering strategy instead emphasizes domain-informed transformations, particularly the use of elemental ratios and nonlinear relationships between chemical components. In forensic analysis, classification decisions are often based on the presence and relative proportion of key elements such as lead, barium,

and antimony, rather than their absolute values. Incorporating these relationships into engineered features is expected to improve classification performance and interpretability.

Model targeted approaches allow us to create discriminative features while reducing noise from less informative variables. The following sections go into detail about each supervised feature engineering plan based on the target model.

3.2.1. Logistic Regression

The feature engineering approach for the logistic regression model was designed with a fundamentally different objective than the methods used for tree-based models and neural networks. While the exploratory data analysis (EDA) highlighted strong non-linear relationships between elemental compositions, as evidenced by the contrast between PCA and UMAP results, logistic regression requires a feature space that supports linear relationships with the log-odds of the target variable. As a result, the focus of this approach was on transforming the data into a stable, scaled, and interpretable representation rather than explicitly modeling complex non-linear interactions.

The primary transformation applied to the raw dataset involved converting elemental concentrations into relative mass-based features. Each elemental measurement was normalized by the total particle mass, producing proportional features that better reflect the compositional nature of the data. This step ensures that the model captures the

relative contribution of each element within a particle, rather than absolute magnitudes, which can vary significantly across observations and introduce unnecessary variance into the model.

Unlike the neural network workflow, which utilized a Centered Log-Ratio (CLR) transformation to address compositional constraints, the logistic regression pipeline intentionally avoided more complex transformations in favor of preserving interpretability. Logistic regression provides direct insight into the relationship between each feature and the probability of classification, and maintaining features in a more intuitive proportional form allows coefficients to be more easily interpreted in terms of elemental influence.

Additional preprocessing steps were implemented to support model stability and convergence. Missing values were handled through imputation to ensure that all observations could be retained during training. Standardization was applied to all numerical features to normalize their scale, which is critical for logistic regression to ensure that coefficient estimates are not disproportionately influenced by feature magnitude. Identifier variables such as `stub_id` and `particle_id` were excluded from the feature set to prevent data leakage and ensure that the model generalizes beyond specific particle groupings.

Feature selection was guided by the need to reduce multicollinearity and improve the reliability of coefficient estimates. Correlation analysis was used to identify redundant

features, particularly among elements that exhibit moderate relationships due to shared compositional constraints. Additional feature distributions were evaluated across GSR and non-GSR classes to assess their ability to provide meaningful separation. Rather than aggressively engineering interaction terms or complex ratios, this approach prioritized retaining features with clear physical meaning, allowing the model to remain interpretable while still capturing relevant signals in the data.

While prior feature engineering efforts for models such as XGBoost emphasized non-linear interactions and elemental ratios these were incorporated more conservatively in the logistic regression framework. The goal was not to maximize model flexibility, but rather to establish a strong, interpretable baseline that could be used for comparison against more complex models. This distinction is important in the context of forensic analysis, where the ability to explain classification decisions is often as critical as predictive performance.

Overall, the feature engineering and selection process for logistic regression was intentionally aligned with the assumptions and strengths of a linear modeling approach. By emphasizing proportional features, standardized inputs, and reduced redundancy, the resulting dataset provides a stable and interpretable foundation for binary classification of GSR versus non-GSR particles.

3.2.2. XGBoost

A key takeaway from our EDA report was that non-linear methods were likely to significantly outperform linear methods due to the complexity of the relationships among the elements in each observed particle. The PCA from the EDA report only captured 16.5% of the variance with heavy overlap between GSR and Non-GSR. In contrast, UMAP revealed distinct clusters with well defined structure, signaling the non-linear nature of the data. The EDA also confirmed the sparsity of the data and the high dimensionality, with only moderate correlations between some elements. These characteristics are all well suited for tree-based models like XGBoost. These models handle sparse data natively by splitting on 0 without needing to impute or transform data, and they exploit the mostly independent signals of the elements by sequential splits. Lastly, in terms of scalability, the large dataset favors the histogram-based splitting of XGBoost which avoids potential training instability that some other non-linear methods might experience.

The method for engineering features for an XGBoost model were based on the key findings from the EDA report. Firstly, the three core GSR elements were likely to be the strongest indicator of a GSR particle. Secondly, we deduced that a comparison of elemental ratios may yield discriminative signals that raw concentrations could not, as the results from PCA were lackluster in comparison to UMAP which strongly suggests complex non-linear relationships. Domain knowledge also supports the idea that elemental ratios can be very informative. One of the core relationships to look at is

barium. The EDA illustrates barium as being problematic for false positives around the GSR/Non-GSR boundaries that overlap. Engineering features to better understand barium's relationship with the other core GSR elements is crucial, but also important may be the relationship between barium and non-GSR elements.

Additional relationships for feature exploration include non-barium confounders, namely environmental or mineral confounders, such as Ca, Si, Al, Fe, Ti, Zn, and Cu. Most of those are grouped in a particle class either with or without barium. In terms of non-barium relationships, zinc and copper (CuZn) were shown to have the strongest correlation as these are found in brass materials, so features that explore this relationship are important. Lastly, oxygen's role in these ratios is curious. Some of the features are explored both with and without oxygen as a part of the total mass. This is because oxygen's presence is so significant, but its informative signal on its own is much less significant. Therefore, there is some possibility that the relationships between the GSR and Non-GSR elements may be more meaningful if oxygen is removed from the ratio.

For a general comparative metric, the PR-AUC score for each standalone feature was estimated using the "average_precision_score" function from the "metrics" module in scikit-learn. PR-AUC directly measures precision-relevant signals, which makes it more useful for our specific study. However, it is important to note that tree-based models don't use single features in a silo, they use them in combination, so while some features

may have a low PR-AUC score they may be critical for learning to distinguish false positives.

3.2.3. Neural Networks

Neural networks learn feature interactions natively, alleviating the need to manually engineer every ratio and interaction, as we did for XGBoost and logistic regression. Instead, we focused on transforming the raw compositional data into a form that removes known statistical artifacts and provides the network with clean, unconstrained inputs.

There were several key design decisions that were made to support how deep learning works under the hood. First, we employed the Centered Log-Ratio (CLR) Transform. Our elemental composition data sums to 100% per particle, also referenced as the "closed composition" problem, which creates spurious correlations and constrains the feature space. The CLR transform (Aitchison, 1986) breaks this constraint by mapping each element to $\log(x_i / \text{geometricMean}(x))$, producing unconstrained real-valued features suitable for neural network input. Without this transform, the data lives on a simplex, not in unconstrained Euclidean space, but NNs assume unconstrained real-valued inputs. The sandbox notebook confirmed this where a multivariate regression of all other elements onto oxygen yielded $R^2 = 0.9494$, but the same test for lead gave $R^2 = 0.9337$. These high R^2 values are artifacts of the sum constraint, not real predictive relationships. The Centered Log-Ratio transform maps each observation from the

simplex to unconstrained real-valued space via $CLR(x_i) = \ln(x_i / g(x))$, where $g(x)$ is the geometric mean of all components. This effectively breaks the sum constraint, preserves relative relationships between elements, produces features centered around zero, and is invertible (we can recover original compositions if needed). Since $\log(0)$ is undefined, we add a small pseudocount before transforming. We use a multiplicative replacement strategy where zeros are replaced with a small value (half the minimum non-zero value per element) and the remaining values are adjusted to maintain the sum constraint.

Second, we decided to use Binary Presence Indicators, consisting of a parallel 27-dimensional binary vector indicating whether each element is present (>0) or absent. This captures the sparsity structure directly and lets the network learn to combine presence patterns with concentration patterns.

Third, we carried only the highest-signal features from the XGBoost exploration forward: top mass-fraction ratio $(Pb + Sb)/(mass - Ba - O)$ (PR-AUC 0.9988), and GSR count (number of core GSR elements present). We keep this set small because the NN learns interactions on its own.

Because NNs are sensitive to the scale of features, especially relevant for numeric stability in gradient based optimization, we used StandardScaler applied after CLR to normalize scale across features.

4. Results

4.1. Preprocessing Decisions and Justifications

The preprocessing pipeline was designed to balance data integrity, model performance, and real-world applicability. The preprocessing pipeline reduced the initial NFI dataset from 2,801,667 particles $\times$ 95 columns to 2,294,985 particles $\times$ 34 columns, containing 27 informative elements + 6 metadata columns + 1 target. The class distribution in the preprocessed dataset is relatively balanced with 47% GSR and 53% non GSR particles, no resampling or class-weight adjustment is anticipated, though class weights will be evaluated during model tuning.

The decision not to move forward with unsupervised methods is detailed extensively in sections 4.6 (PCA) and 4.7 (UMAP) of the EDA report, which include visuals, statistics, and interpreted analysis to support the decision. The PCA section illustrates the considerable lack of explained variance, capturing only 16.5% and showing heavy overlap between GSR and Non-GSR in the reduced space. And while UMAP helped confirm that non-linear relationships were predominant, we feel that elemental weight percentages and nonlinear relationships across particle ratios will be primary drivers for the ML model. Statistical clustering based on abstract embedding will likely fail to consider important domain knowledge from areas like chemistry and forensics. Such domain knowledge can be pertinent when determining why a particle is classified as GSR, especially in legal situations.

4.2. Feature Engineering Findings

4.2.1. Logistic Regression

The feature engineering approach for logistic regression produced a structured and domain-informed feature space designed to compensate for the model's inability to learn nonlinear relationships and interactions natively. The final engineered dataset includes transformed elemental features, ratio-based features, compositional percentages, interaction terms, and domain-driven summary metrics, all constructed without introducing missing or invalid values.

The application of logarithmic transformations to key elemental concentrations (e.g., Pb, Ba, Sb) significantly reduced the extreme right-skew observed in the raw data. Prior to transformation, elemental distributions were heavily concentrated near zero with long tails of high values. After applying $\log_{10}$ transformations, the distributions became more compressed and approximately linear in shape, improving their suitability for a linear model. This transformation enhances numerical stability and allows logistic regression to better approximate decision boundaries in the transformed feature space.

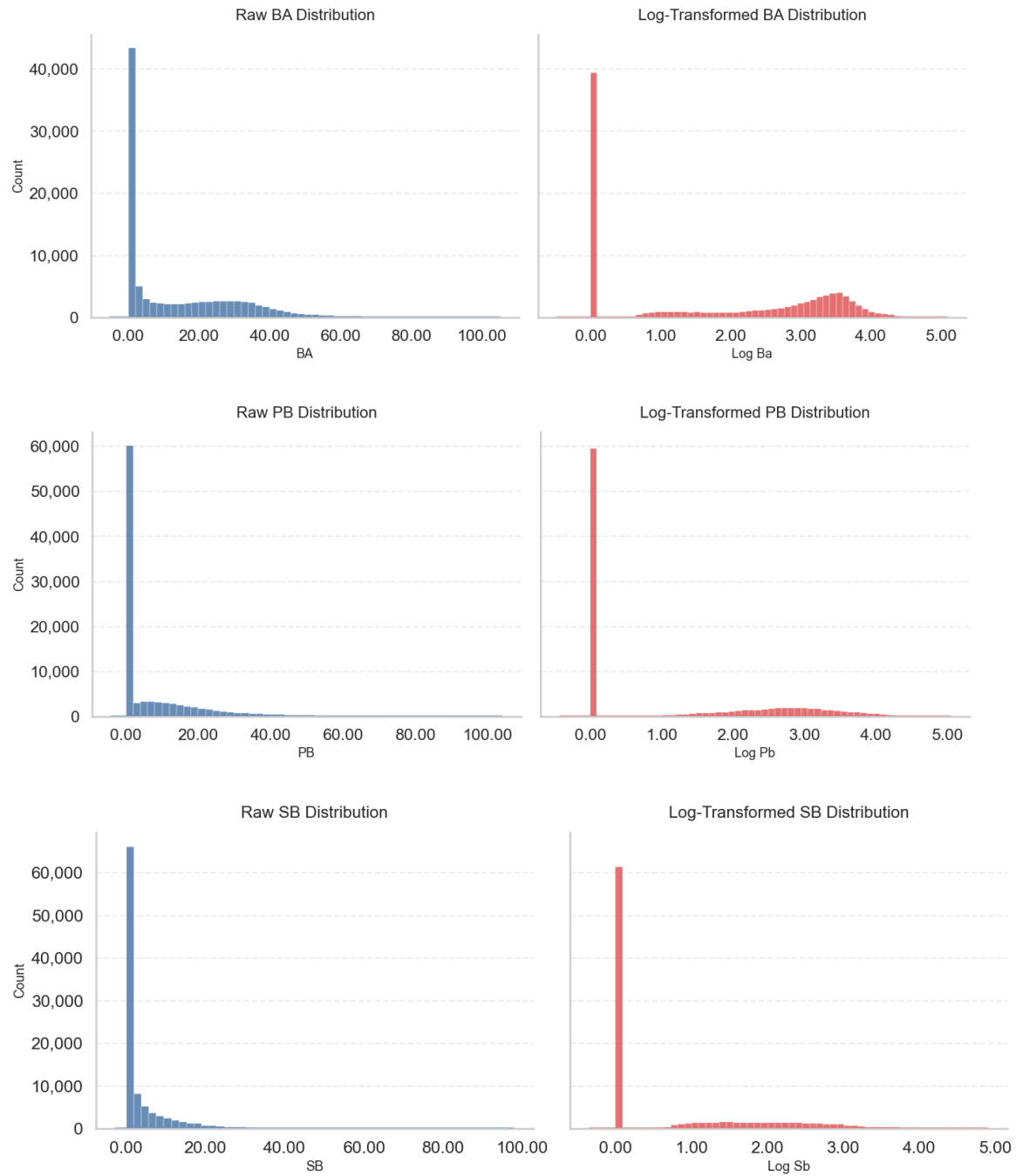


Fig. 1: Distribution of Key Elements Before and After Log Transformation

Ratio-based features emerged as one of the most important components of the engineered feature set. Ratios among core GSR elements (Pb/Ba, Pb/Sb, Ba/Sb) and between GSR elements and environmental confounders (e.g., Pb/Zn, Sb/Cu, Ba/Ti) were well-behaved and did not produce extreme or unstable values due to the use of small epsilon adjustments in denominators. These features capture relative chemical relationships that are critical for distinguishing true GSR particles from non-GSR particles with similar individual elemental signatures. This aligns with forensic principles, where classification decisions rely on proportional composition rather than absolute concentrations.

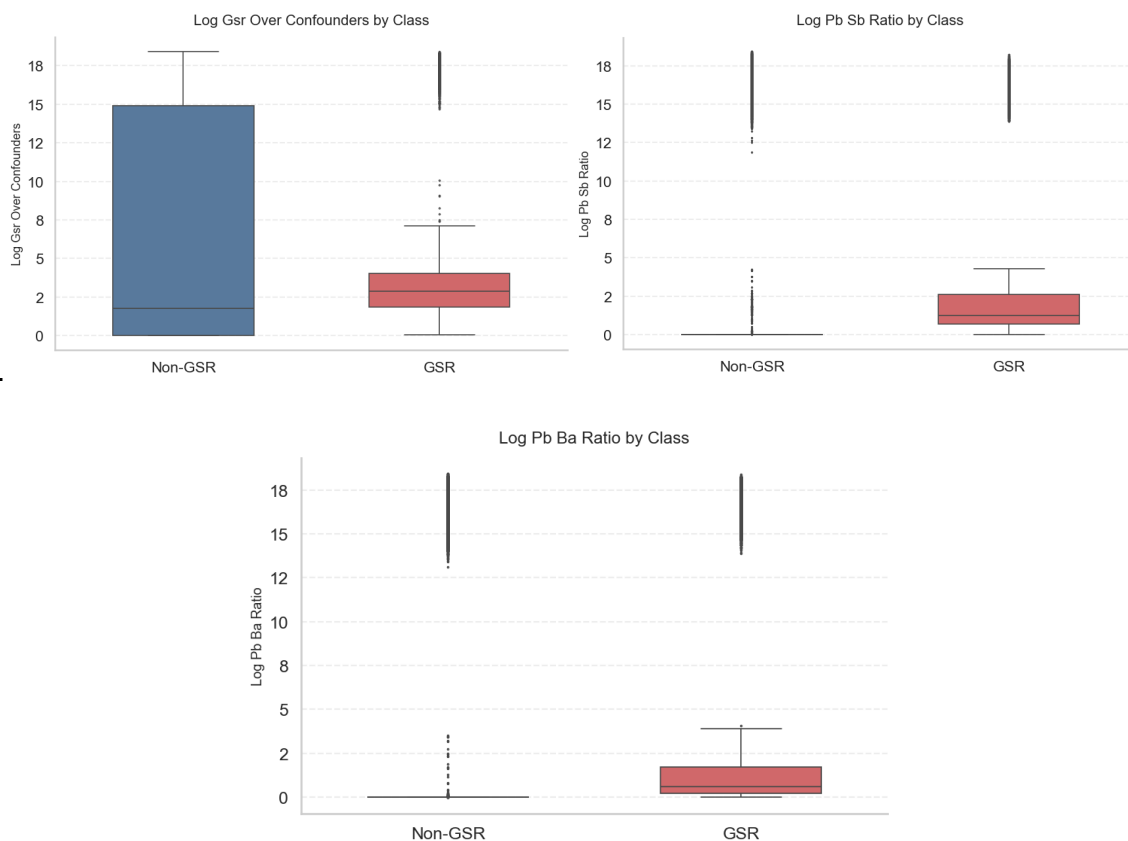


Fig. 2: Distribution of Engineered Ratio Features (Pb/Ba, Pb/SB, etc.)

The compositional percentage features (element mass divided by total particle mass) provided a normalized representation of each particle's elemental makeup. These features were consistently bounded between 0 and 1 and did not introduce instability into the dataset. While they do not fully eliminate the closed-composition constraint, they improve interpretability and ensure comparability across particles of varying total mass.

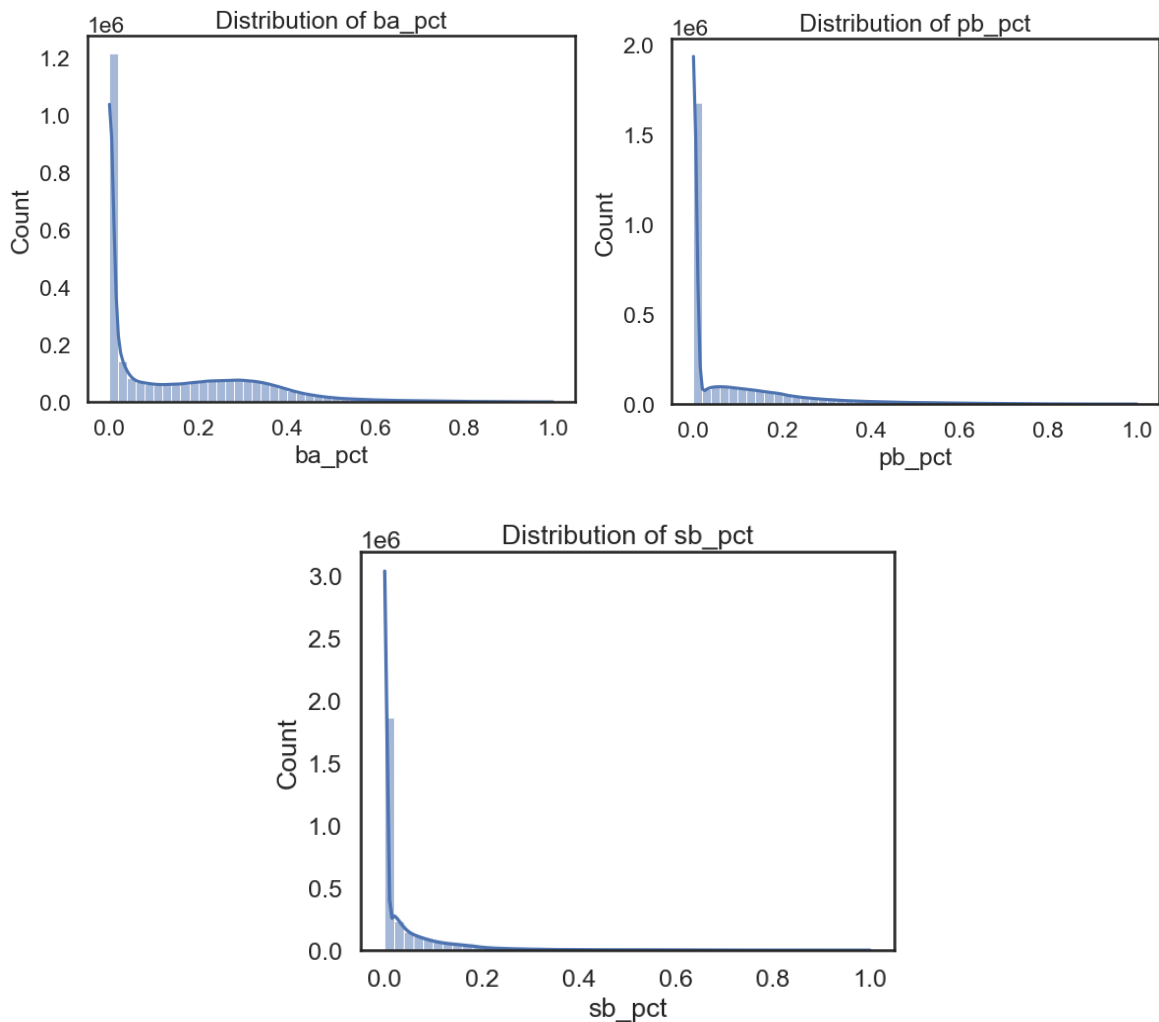


Fig. 3: Distribution of of Elemental Percentage Features (Pb, Ba, Sb)

Interaction terms between key elements ($\text{Pb} \times \text{Sb}$, $\text{Ba} \times \text{Sb}$, $\text{Pb} \times \text{Ba}$) were successfully generated and provided additional nonlinear signal to the model. These features are particularly important given the overlapping class boundaries observed during UMAP analysis, where combinations of elements—not individual values—drive separation between GSR and non-GSR classes. By explicitly including these interactions, the

logistic regression model can approximate more complex relationships that would otherwise be inaccessible in a purely linear framework.

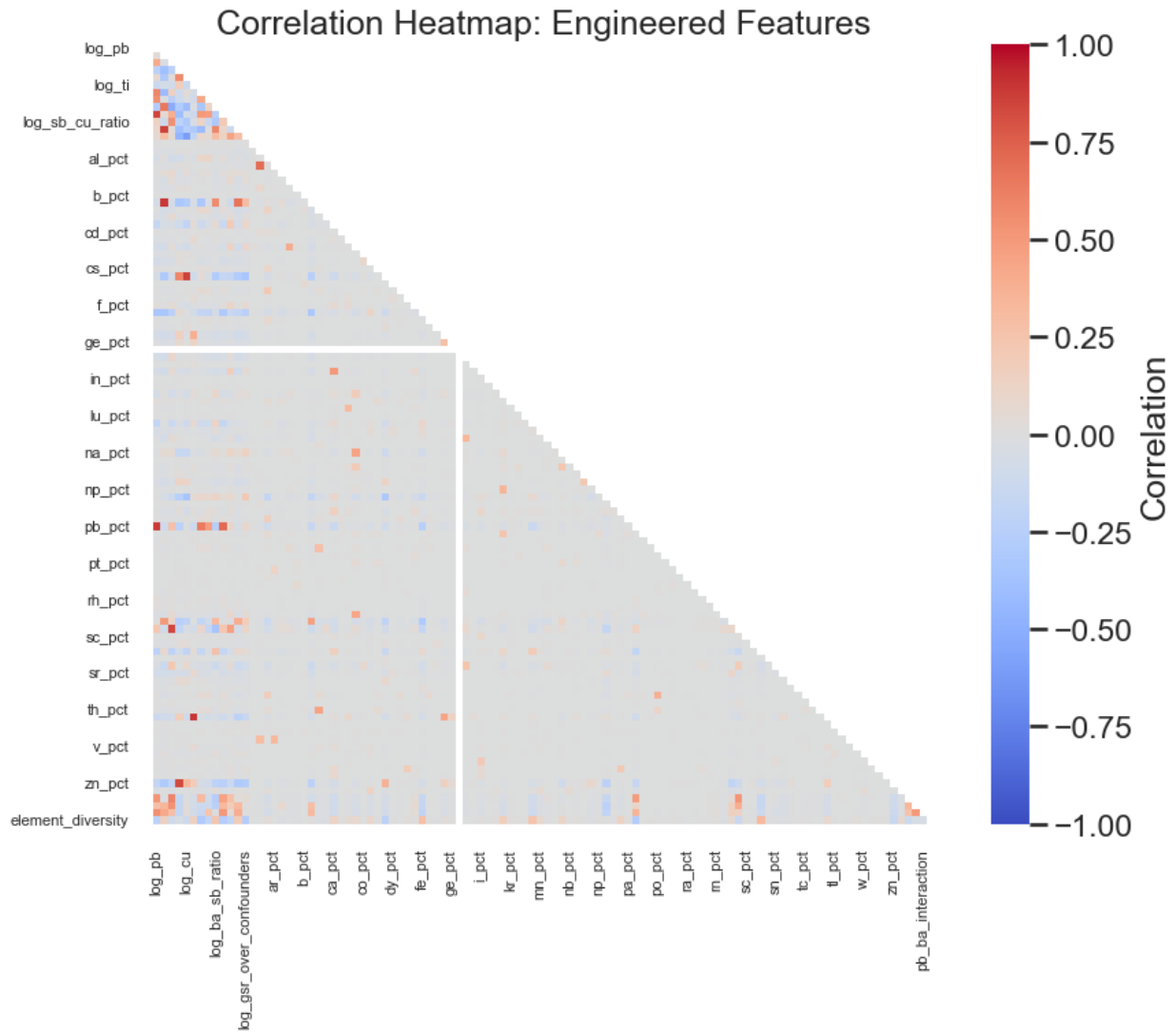


Fig. 4: Correlation Heatmap: Engineered Features

Domain-driven summary features, including the ratio of total GSR elements to confounders and the element diversity count, provided additional high-level signal. The

element diversity feature showed meaningful variation across particles, reflecting differences in compositional complexity, while the GSR-to-confounder ratio captured overall signal strength relative to environmental noise.

No NaN or infinite values were introduced during feature engineering, confirming that all transformations were numerically stable. Missingness checks across engineered features showed negligible or zero missing values, and all features were suitable for direct use in modeling after scaling.

The class distribution remained stable at approximately 47% GSR and 53% non-GSR following feature engineering, indicating that no bias was introduced during transformation. Additionally, the preservation of the `stub_id` identifier enables group-based splitting to prevent leakage, ensuring that particles from the same stub do not appear in both training and testing datasets.

Overall, the feature engineering process for logistic regression successfully transformed the raw elemental data into a richer, more expressive feature space. The combination of log transformations, ratios, compositional features, and interaction terms provides the model with the necessary structure to approximate nonlinear relationships while maintaining interpretability and alignment with forensic domain knowledge.

4.2.2. XGBoost

The feature exploration for XGBoost resulted in 14 different potential features. Of these 14, there are 10 features that are paired features where only 1 of the 2 will be used. One version uses oxygen in the total mass calculation and the other removes oxygen from the total mass calculation. As mentioned in the “Methods” section, this was out of curiosity to see if oxygen diluted any relationships. Ultimately, it is expected that 9 of the 14 engineered features will be selected for the XGBoost model.

Table 1: PR-AUC Estimates by Feature

	NewFeature	PR-AUC estimate
1	pb_sb_over_non_ba_o_mass	0.998800
2	pb_sb_over_non_ba_mass	0.997900
3	core_gsr_count	0.997600
4	log_pb_plus_sb	0.997300
5	gsr_over_confounders	0.943000
6	pb_times_sb	0.832300
7	pb_ba_over_non_sb_o_mass	0.773600
8	pb_ba_over_non_sb_mass	0.692500
9	ba_sb_over_non_pb_o_mass	0.691400
10	ba_sb_over_non_pb_mass	0.617200
11	cu_zn_over_mass	0.432900
12	cu_zn_over_non_o_mass	0.429800
13	ti_zn_over_mass	0.401000
14	ti_zn_over_non_o_mass	0.400400

Intuitively, the first feature that was engineered was the count of core GSR elements existing in the particle. This is the most straightforward feature because it is simply implementing a boolean for Pb>0, a boolean for Ba>0, and a boolean for Sb>0. Any of

those that are True counts as 1, therefore a maximum of count 3 means that all core GSR particles are present in the particle. Overall this feature is very strong when all 3 core elements are present or when 0 to 1 core elements are present. However, it is important to note two things. Firstly, the ambiguous labeled particles have been removed which may be why it seems when 1 core element is present this feature is accurate. In reality, it is possible that this is not accurate. Secondly, this feature falls short when trace amounts of these core elements are present because it does not consider quantities, only whether it exists or not.

Table 2: Target Distribution by Core GSR Count

core_gsr_count	0	1	2	3	All
label					
GSR	0	0	536881	542065	1078946
Non_GSR	484704	726207	5092	36	1216039
All	484704	726207	541973	542101	2294985

As seen in the table above, the biggest concern is the 36 Non-GSR particles with 3 core elements present, as these have high-risk for false positives. The first step is to analyze the elemental composition of these particles.

Additionally, understanding which particle classes these 36 particles belong to will help identify next steps for a new feature to catch them.

Table 3: Element Composition & Classes for 36 Non-GSR Particles w/ 3 Core GSR

median	
o	46.182249
ba	22.044205
al	4.948296
pb	0.956436
sb	0.771331
s	0.722646
cu	0.635611
si	0.507025
ca	0.309733

final_class	
BaAl	30
CuZn	5
BaCaSi	1

From these numbers, it becomes clear that there may be some contamination of lead (Pb) and antimony (Sb) in these particles, as trace amounts are shown in BaAl, CuZn, and BaCaSi, all of which are strong Non-GSR classes. To address these edge cases, a new feature that quantifies Pb and Sb would complement the “Core GSR Count” feature. Specifically, multiplicative quantification would properly address these scenarios:

1. Large quantities of both Pb and Sb are weighted higher.
2. Trace quantities of both Pb and Sb are penalized strongly.
3. A large quantity of 1 and trace quantity of the other will be slightly penalized.

The reason to penalize the third scenario, as opposed to an additive feature (Pb + Sb), is because a trace amount of any of these elements should be considered as potential

contamination and weighted accordingly to minimize false-positives. Ultimately, this is why the **Pb x Sb** feature was created (pb_times_sb).

Next, we considered the 5,092 Non-GSR particles with only 2 core GSR elements. While the multiplicative quantification works well when all 3 core GSR elements are present, when Core GSR Count = 2 there is a risk that either Pb or Sb is absent and therefore 0, which means the Pb x Sb value is 0 even if lead and barium levels are very high or barium and antimony levels are very high. This is not ideal. To address this, an alternative *additive* quantification method is the **log_pb_plus_sb** feature. This logarithmic approach puts more separation between the lower PbSb quantities and the higher PbSb quantities.

We needed to further analyze these 5,092 particles because we can't assume they're all related to trace amounts of lead and antimony. We found that in addition to the barium and brass classes, there are several new particle classes that appear not to be related to contamination, as seen in the table below.

Table 4: Class Distribution & Pb Sb Composition for 5,092 Non-GSR Particles w/ 2 Core GSR

	count	pb_median	pb_max	sb_median	sb_max
final_class					
BaAl	4077	0.000000	11.620400	0.803285	1.978046
BaCaSi	661	0.942170	7.834150	0.000000	0.995508
CuZn	298	0.000000	21.747881	1.624879	3.569439
Hg	35	0.760992	2.336872	20.959925	56.606514
TiZnGd	18	17.198016	73.943672	3.270193	11.296607
ZnTi	3	7.850978	9.904734	0.961876	0.971031

Initially, features were explored that directly compared ratios of elements (e.g. Pb/Ba). However, this results in scenarios where the element in the denominator may not be present and therefore the ratio is undefined. In this case, if the element in the numerator is present, the result is infinity, whereas if the element in the numerator is present then the result is NaN. In either case, the ratio is not a measurable number. There are several methods for addressing this, but at the risk of accidentally introducing bias the decision was made to compare element ratios over the total mass of the particle.

Naturally, the relationship between the core GSR elements is important to incorporate. These features compare the additive quantity of 2 of the core GSR elements over the particle's mass *without* the 3rd particle:

- $(\text{Pb} + \text{Ba}) / (\text{particle mass} - \text{Sb})$
- $(\text{Pb} + \text{Sb}) / (\text{particle mass} - \text{Ba})$
- $(\text{Ba} + \text{Sb}) / (\text{particle mass} - \text{Pb})$

Then, based on the particle class distribution of the previous table, additional features that address brass particles (CuZn) and the TiZn relationship are needed:

- $(\text{Cu} + \text{Zn}) / \text{particle mass}$
- $(\text{Ti} + \text{Zn}) / \text{particle mass}$

All five of these features were also calculated with oxygen removed from the particle mass as well to assess whether oxygen might dilute the relationships.

The last feature considers environmental and mineral confounders in relationship to lead and antimony, while intentionally removing barium from the equation knowing that barium is problematic as it is heavily present in both GSR and Non-GSR particles.

Based on domain knowledge and EDA, the most common confounders to consider are Ca, Si, Al, Fe, Ti, Zn, and Cu. This **gsr_over_confounders** feature simply takes Pb+Sb and divides by the sum of the confounders. However, similar to when doing direct element ratios, there is a risk that the denominator will be 0 and therefore the ratio either NaN or infinity.

To address this specific case, it was decided to use a sentinel value strategy with the expectation that the XGBoost tree model will identify these values as unreliable. The sentinel value of **-1** is used to ensure that any instance of NaN or infinity results in a negative number, and therefore the model should learn that negative ratios are unreliable and the feature should only be used when the ratio is positive. The reason for not using an epsilon approach here is that when analyzing the data, there were 196

Non-GSR particles whose ratio evaluated to infinity and therefore would have had a very high value. This feature logically should have high-value ratios corresponding to GSR particles (Pb + Sb much higher than sum of confounders), and therefore having Non-GSR particles with high-value ratios creates risk for false positives.

4.2.3. Neural Network

The CLR transform was applied to the 27 informative elements for all ~2M particles. Prior to transformation, the zero rate across all element-particle entries was 70.98%, confirming the extreme sparsity identified during EDA. After multiplicative replacement, the zero rate dropped to 0%. After the CLR transform, all 27 output features are real-valued and unconstrained, with row sums effectively zero (order 10^{-14}), confirming the transform was applied correctly.

The correlation structure before and after CLR was compared using heatmaps on a random 50,000-particle sample, accommodating local compute constraints. The raw correlation matrix is dominated by strong negative correlations between oxygen and nearly every other element, an artifact of the closed-composition constraint, when oxygen is high, everything else must be low, regardless of any real chemical relationship. After CLR transformation, this artifact is substantially reduced, and the remaining correlations reflect genuine chemical co-occurrence patterns rather than mathematical constraints. For example, the Pb-Sb positive association (both are core GSR elements that co-occur in gunshot residue) becomes more clearly visible in the

CLR-transformed space, while the spurious O-vs-everything negative pattern is attenuated.

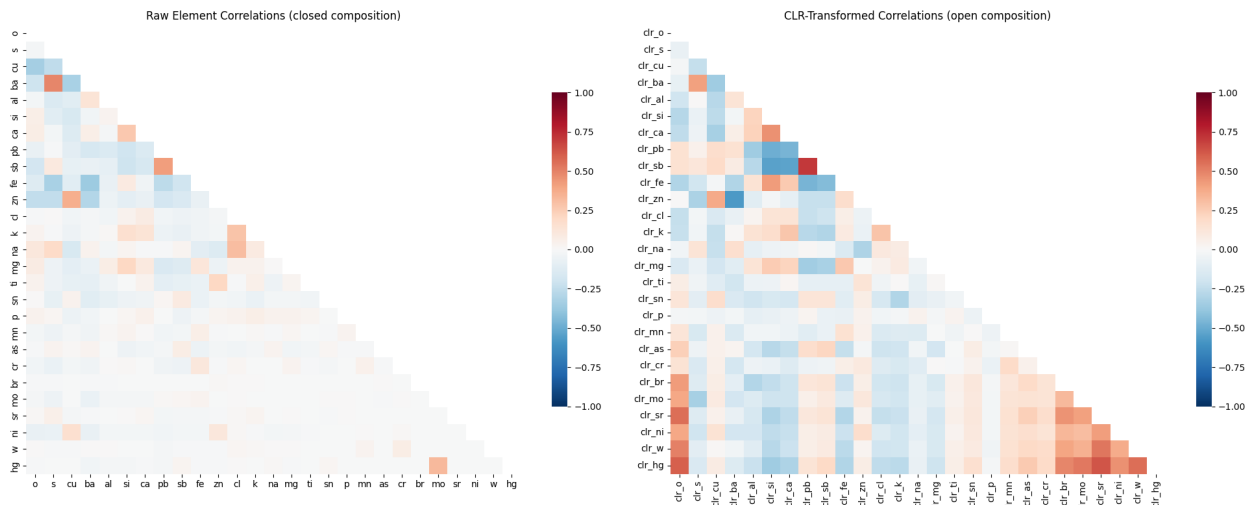


Fig. 5: Correlation comparison between raw and CLR-transformed element correlations

The 27 binary presence flags capture the sparsity structure of the data. The top presence rates are: O (99.8%), S (80.3%), Ba (70.0%), Cu (67.7%), Al (59.4%), Si (55.0%), Ca (44.8%), Pb (42.4%), Zn (41.0%), and Sb (37.3%). Notably, the three core GSR elements have presence rates of Ba = 70.0%, Pb = 42.4%, and Sb = 37.3%. The fact that Pb and Sb are absent in the majority of particles means that their mere presence carries strong signal in this particular dataset, captured by the GSR count engineered feature.

The three engineered features were computed without any NaN or Inf values in the final output. The mass-fraction ratio *pb_sb_over_non_ba_o_mass* has a mean of 0.24 and

standard deviation of 0.31, with values ranging from 0 to 1, indicating it is well-behaved and does not require additional clipping or transformation. The *log_pb_plus_sb* feature has a mean of 1.36 and standard deviation of 1.54, ranging from 0 to 4.62. The GS count feature has a mean of 1.50 and median of 1.0, with values in {0, 1, 2, 3}, and the 25th percentile is 1, meaning most particles contain at least one of the three core GSR elements.

The distribution comparison between raw and CLR-transformed concentrations for the three core GSR elements (Pb, Ba, Sb) reveals that the CLR transform produces approximately symmetric, bell-shaped distributions centered near zero, in contrast to the heavily right-skewed raw distributions, as well as showing better class separation. This is favorable for neural network training because gradient-based optimizers perform better when input features have roughly symmetric distributions without extreme skew. The visible separation between GSR and non GSR classes, particularly for Pb and Sb, suggests the transform potentially enhances the discriminative signal.

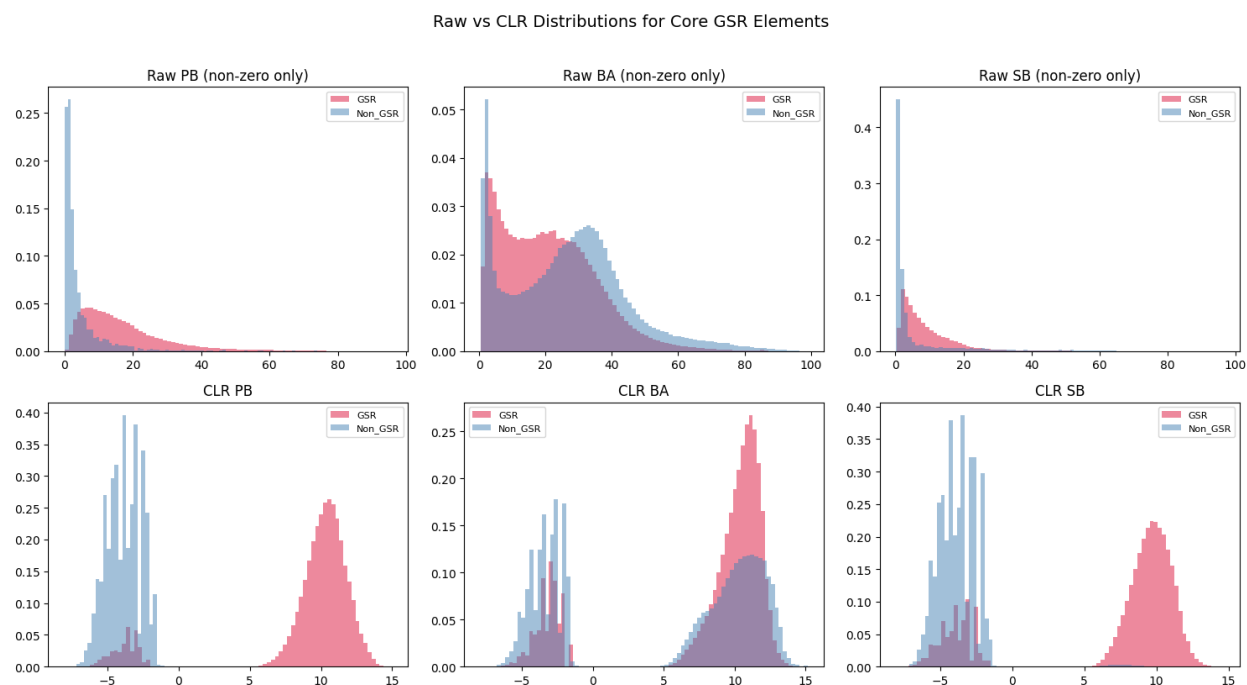


Fig. 6: Comparison of raw and CLR transformed distributions

The assembled neural network feature matrix contains 2,294,985 particles $\times$ 57 features, with zero NaN and zero Inf values. The preprocessing and feature engineering work for the neural network produced a principled 57-feature representation that addresses the key statistical challenges identified during EDA. The most significant finding was the closed-composition problem. Because elemental percentages sum to 100% per particle, raw concentrations contain spurious correlations that can mislead models. The CLR transform resolves this by mapping the data from the constrained simplex to unconstrained Euclidean space, and the before/after correlation comparison confirms that the dominant artifact (the negative O-vs-everything pattern) is substantially

reduced after transformation. This was not part of our original analysis plan and represents a meaningful revision informed by the EDA and sandbox exploration.

The feature engineering philosophy for the NN track is deliberately minimal compared to the XGBoost and logistic regression tracks. Where XGBoost requires explicit ratio and interaction features because trees split on one variable at a time, and logistic regression requires log transforms to approximate non-linear boundaries, the neural network learns arbitrary non-linear mappings through its hidden layers. The 57-feature set balances giving the network clean inputs (CLR), structural information (presence flags), and a small number of high-signal domain features (the three engineered features) without prescribing how those inputs should be combined.

Referring back to our research question, the preprocessing results are encouraging. The 47/53 class balance means the binary classification task is well-posed without requiring aggressive resampling methods that can introduce bias such as SMOTE. The CLR-transformed features show visible separation between GSR and non GSR classes for the core elements, and the top engineered features from the XGBoost exploration (PR-AUC > 0.99) suggest strong discriminative signal exists in the data. The question is whether the neural network can exploit the full 27-dimensional compositional space more effectively than the hand-engineered features alone.

5. Discussion & Next Steps

Our preprocessing and feature engineering produces three distinct and model specific feature sets that each address modeling requirements and challenges identified in the EDA. The most consequential finding was the closed-composition constraint, which was resolved with the CLR transform producing visible separation and approximate symmetry in the distributions of core GSR elements. For XGBoost, the mass-fraction ratio features proved discriminative and showed strong signal based on PR-AUC scores. For logistic regression, the log transform successfully reduced right skew of the raw elemental distributions and ratio based features capture the compositional relationships relevant for our modeling goal. We also show solid balance between GSR and non GSR particles, reducing the need for resampling methods like SMOTE which can introduce artifacts that can become problematic for modeling such as artificial bias injection.

The next phase of the project will focus on developing and evaluating predictive models using the engineered features. Initial modeling will include logistic regression as a baseline approach, followed by more flexible methods such as gradient boosted and deep learning models.

Model performance will be evaluated using metrics such as ROC-AUC, confusion matrices, and classification reports. Particular attention will be given to performance across all three classes, including the ambiguous category.

Feature importance techniques will be used to refine the feature set and identify the most predictive variables. Additionally, engineered features such as elemental ratios and nonlinear transformations will be incorporated into the modeling pipeline to improve performance.

All models will be validated using group-based cross-validation to ensure consistency with the preprocessing strategy and to prevent data leakage. This approach will provide a more accurate assessment of how well the model generalizes to new, unseen data.

Ultimately, the goal is to develop a model that not only achieves strong predictive performance but also provides interpretable results that align with forensic principles. This is particularly important given the legal implications of GSR classification, where transparency and explainability are essential.

The following sections give a short overview of the exact steps planned for each of our 3 models.

5.1 Logistic Regression

The next phase of the logistic regression track will focus on developing a robust baseline classification model using the engineered feature sets. Logistic regression serves as an important benchmark due to its interpretability and its ability to provide insight into which features drive classification decisions, which is particularly valuable in forensic and legal contexts.

Model development will begin with fitting a logistic regression model using Scikit-learn, implemented within a pipeline that includes imputation (if necessary) and StandardScaler to normalize feature magnitudes. Scaling is critical for logistic regression because the optimization process is sensitive to differences in feature scale, particularly when regularization is applied.

Training and evaluation will be conducted using a group-based train/test split based on stub_id, consistent with the preprocessing strategy. This ensures that all particles from a given stub are contained entirely within either the training or testing set, preventing leakage of stub-specific patterns and providing a more realistic estimate of model generalization performance.

Multiple feature sets will be evaluated, including the full engineered feature set, an engineered-only subset, and reduced feature sets that exclude dominant GSR indicators. This approach allows for controlled experimentation to assess the marginal contribution of different feature types and to determine whether simpler, more interpretable models can achieve comparable performance.

Model performance will be evaluated using a comprehensive set of metrics, including accuracy, precision, recall, F1-score, and ROC-AUC. Particular emphasis will be placed on minimizing false positives, where non-GSR particles are incorrectly classified as GSR, as these errors have significant implications in forensic investigations. Confusion

matrices will be analyzed in detail, including breakdowns by particle subclass where possible, to identify systematic failure modes.

To address potential overfitting and improve generalization, regularization techniques (L1 and L2 penalties) will be explored. Hyperparameter tuning will focus on the regularization strength (C parameter) and penalty type, with cross-validation used to identify optimal configurations.

Feature importance and interpretability will be central to this phase. Model coefficients will be examined to understand the direction and magnitude of each feature's influence on the probability of GSR classification. This analysis will be used to validate whether the model aligns with known forensic principles, such as the importance of Pb, Ba, Sb, and their relative proportions. Additional techniques such as permutation importance may be used to further assess feature contributions.

Finally, results from the logistic regression model will be compared against more flexible models, including XGBoost and neural networks, to evaluate the trade-off between interpretability and predictive performance. While logistic regression is not expected to outperform nonlinear models, it provides a transparent and defensible baseline that is essential in high-stakes applications such as forensic evidence analysis.

Overall, the logistic regression modeling phase will focus on building an interpretable, well-calibrated baseline model, validating the effectiveness of the engineered features, and establishing a foundation for comparison with more complex modeling approaches.

5.2 XGBoost

After carefully considering leakage and bias concerns and finalizing selected features, an appropriate ratio will be determined for the train-test split, which as previously mentioned will utilize the GroupShuffleSplit to ensure no stub ID leakage. A baseline XGBoost model will be established using moderate hyperparameters to get an idea of performance. Then, evaluation techniques for precision, recall, and false-positives will be checked along with hyperparameter tuning using techniques such as RandomizedSearchCV with StratifiedGroupKFold. The default threshold will be optimized, as 0.5 will likely not be suitable for this model. Lastly, SHAP analysis will be computed using TreeExplainer on a sample from the test set. This will determine which features drive predictions. Isolating false-positive particles in the SHAP sample can also be done to understand and minimize occurrences.

5.3 Neural Network

The modeling phase will focus on a fully connected neural network (multi-layer perceptron) on the 57-feature input. The planned architecture is a small MLP with 2 to 3 hidden layers of 64 to 128 units each, using ReLU activations and dropout for regularization. StandardScaler will be applied within the training pipeline (fit on training data, transform on test data) to normalize the CLR and engineered features. Binary cross-entropy loss will be used with the Adam optimizer. Hyperparameter tuning will focus on hidden layer size, dropout rate, learning rate, and batch size, with evaluation

oriented toward minimizing false positives (non GSR particles incorrectly classified as GSR) while maintaining acceptable recall. We will evaluate using accuracy, precision, recall, F1, ROC-AUC, and PR-AUC, with confusion matrices broken down by particle subclass (e.g., PbBaSb, BaAl, CuZn) to identify where the model succeeds and fails. SHAP values will be computed on the trained model to assess which features and elements drive predictions, connecting back to our interpretability research question.

5.4 One final Note on Train/Test/Val Splits

Rather than performing our train-test-validation split in the preprocessing pipelines, we reserve it for the modeling phase. All of our feature transformations are computed on a per-particle basis and do not use any cross-observation statistics that could introduce data leakage. The CLR transform computes the geometric mean across the 27 elements within a single particle's own row, not across the dataset. The binary presence flags and the mass-fraction ratio are all deterministic per-row arithmetic operations that reference only that particle's own elemental concentrations. Because no transform in this pipeline learns population-level parameters from the data, the output is identical regardless of whether the split happens before or after. The one operation that touches multiple rows is the multiplicative replacement pseudocount (half the global minimum non-zero value), but this is a single scalar constant on the order of 10^{-6} that would be virtually identical whether computed on the full dataset or a training subset. Transforms that do depend on population statistics, specifically StandardScaler normalization, are

deferred to the modeling pipeline where they will be fit exclusively on training data and applied to the test and validation sets, preventing any information leakage.

Citations

Citations

Matzen, T., Kukurin, C., van de Wetering, J., Ariëns, S., Bosma, W., Knijnenberg, A., ... & Ypma, R. J. (2022). Objectifying evidence evaluation for gunshot residue comparisons using machine learning on criminal case data. *Forensic science international*, 335, 111293.

Ritchie, Nicholas W. M., Renolds, Amy (2021), Automated particle analysis (SEM/EDS) data from samples known to have been exposed to gunshot residue and from samples occasionally mistaken for gunshot residue - like brake dust and fireworks., National Institute of Standards and Technology, <https://doi.org/10.18434/mds2-2476> (Version: 1.0, Accessed 2026-03-21)

Ritchie NWM. Reproducible Spectrum and Hyperspectrum Data Analysis Using NeXL. *Microsc Microanal*. 2022 Mar 2:1-18. doi: 10.1017/S143192762200023X. Epub ahead of print. PMID: 35232526; PMCID: PMC9437143.
<https://pmc.ncbi.nlm.nih.gov/articles/PMC9437143/#R24>

Appendices

Appendix A: GitHub Repository (*Team Delta Capstone*)

<https://github.com/bkoconnell/datascience-capstone>

Preprocessed data file → **data/processed/preprocessed.parquet**

Feature Processing Notebooks → **notebooks/02_feature_processing**

Appendix B: EDA Data Dictionary

https://github.com/bkoconnell/datascience-capstone/tree/main/artifacts/reports/01_eda/data_dictionaries/eda_nfi_particle_labeled.md

Appendix C: Data Dictionary Neural Network

The neural network feature dataset (engineered_features_nn.parquet) contains 2,294,985 rows and 62 columns.

Metadata Columns

Column	Type	Description
stub_id	int	SEM stub identifier
particle_id	int	Particle identifier within stub
label	string	"GSR" or "Non_GSR"
target	int	Binary target: GSR = 1, Non_GSR = 0

final_class	string	Particle subclass (e.g., PbBaSb, CuZn, BaAl)
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CLR-Transformed Element Features (27 columns)

Each column is the Centered Log-Ratio transform of the corresponding element's weight-percent concentration. Values are real-valued and unconstrained (can be negative). Naming convention: *clr_{element}* .

Column	Element
clr_o	Oxygen
clr_cl	Chlorine
clr_cr	Chromium
clr_s	Sulfur
clr_k	Potassium
clr_br	Bromine
clr_cu	Copper
clr_na	Sodium
clr_mo	Molybdenum
clr_ba	Barium
clr_mg	Magnesium
clr_sr	Strontium
clr_al	Aluminum
clr_ti	Titanium
clr_ni	Nickel

clr_si	Silicon
clr_sn	Tin
clr_w	Tungsten
clr_ca	Calcium
clr_p	Phosphorus
clr_hg	Mercury
clr_pb	Lead
clr_mn	Manganese
clr_sb	Antimony
clr_as	Arsenic
clr_fe	Iron

Binary Presence Features (27 columns)

Each column is 1 if the corresponding element has concentration > 0, and 0 otherwise.

Naming convention: *has_{element}* .

Engineered Features (3 columns)

Column	Formula	Description
pb_sb_over_non_ba_o_mass	$(\text{Pb} + \text{Sb}) / (\text{total_mass} - \text{Ba} - \text{O})$	Mass fraction of Pb+Sb excluding Ba and O; sentinel value -1 when denominator = 0
log_pb_plus_sb	$\log_{10}(\text{Pb} + \text{Sb})$	Log-transformed additive GSR signal
gsr_count	count of {Pb, Ba, Sb} > 0	Number of core GSR elements present (0–3)

Appendix D: Data Dictionary Logistic Regression

Metadata Columns

Column	Type	Description
stub_id	int	SEM stub identifier
particle_id	int	Particle identifier within stub
label	string	"GSR", "Non_GSR", or "Ambiguous"
target	int	Binary target: GSR = 1, Non_GSR = 0

Log-Transformed Element Features

Column	Description
log_pb	Log-transformed lead concentration
log_ba	Log-transformed barium concentration
log_sb	Log-transformed antimony concentration

Ratio Features

Column	Formula	Description
log_pb_sb_ratio	$\log(\text{Pb} / \text{Sb})$	Relative relationship between Pb and Sb
log_pb_ba_ratio	$\log(\text{Pb} / \text{Ba})$	Relative relationship between Pb and Ba
log_ba_sb_ratio	$\log(\text{Ba} / \text{Sb})$	Relative relationship between Ba and Sb

GSR vs Confounder Features

Column	Formula	Description
log_pb_zn_ratio	$\log(\text{Pb} / \text{Zn})$	Distinguishes GSR from environmental particles
log_sb_cu_ratio	$\log(\text{Sb} / \text{Cu})$	Separates GSR from copper-rich particles
log_ba_ti_ratio	$\log(\text{Ba} / \text{Ti})$	Differentiates industrial/environmental sources
log_gsr_over_confounders	Combined ratio	Overall GSR signal vs confounders

Composition Features

Column	Description
pb_pct	% of particle mass from Pb
ba_pct	% of particle mass from Ba
sb_pct	% of particle mass from Sb

Interaction Features

Column	Description
pb_sb_interaction	Interaction between Pb and Sb
pb_ba_interaction	Interaction between Pb and Ba
ba_sb_interaction	Interaction between Ba and Sb